

RaceDay_API_Endpoint_Plan

TABLE: AUTHENTICATION ENDPOINTS					
HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/auth/register	Register a new user account with role selection	None (Public)	{ "email": "string", "password": "string", "firstName": "string", "lastName": "string", "dateOfBirth": "2026-01-01", "role": "Participant" }	201 Created - { "user": { "id", "email", "firstName", "lastName", "role" }, "message": "Registration successful" } 400 Bad Request - validation errors 409 Conflict - "Email already registered"
POST	/api/auth/login	Authenticate user and return JWT token	None (Public)	{ "email": "string", "password": "string" }	200 OK - { "token": "jwt_token", "user": { "id", "email", "firstName", "lastName", "role" } } 401 Unauthorized - "Invalid email or password"

USER PROFILE ENDPOINTS					
HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/users/profile	Get current user's profile	Any (Logged in)	None	200 OK - { "id", "email", "firstName", "lastName", "dateOfBirth", "profileImageUrl", "role", "createdAt" } 401 Unauthorized - "Not authenticated"
PUT	/api/users/profile	Update current user's profile	Any (Logged in)	{ "firstName": "string", "lastName": "string", "dateOfBirth": "2026-01-01", "profileImageUrl": "string" }	200 OK - updated user profile 400 Bad Request - validation errors
GET	/api/users/{id}	Get user by ID	Organiser	None	200 OK - user details 403 Forbidden - "Insufficient permissions" 404 Not Found - "User not found"

EVENT ENDPOINTS					
HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/events	Get all events with optional filtering	None (Public)	None (Query: ?status=upcoming&search=keyword&page=1&pageSize=10)	200 OK - { "items": [events], "totalCount": 50, "page": 1, "pageSize": 10, "totalPages": 5 }
GET	/api/events/{id}	Get specific event by ID with categories	None (Public)	None	200 OK - event details with organiser and categories 404 Not Found - "Event not found"
POST	/api/events	Create a new event	Organiser	{ "eventName": "string", "description": "string", "eventDate": "2026-01-01T08:00:00", "location": "string", "venue": "string", "distanceKm": 10.0, "routeMapURL": "string", "categories": [{ "categoryId": 1, "entryFee": 150.00, "maxParticipants": 500 }] }	201 Created - event details with categories 400 Bad Request - validation errors 403 Forbidden - "User must be an

					organiser "
PUT	/api/events/{id}	Update an existing event	Organi ser	{ "eventName": "string", "description": "string", "eventDate": "2026-01-01T08:00:00", "location": "string", "venue": "string", "distanceKm": 10.0, "routeMapURL": "string", "isActive": true }	200 OK - updated event 403 Forbidde n - "Not the event organiser " 404 Not Found - "Event not found"
DELE TE	/api/events/{id}	Delete an event	Organi ser	None	204 No Content 403 Forbidde n - "Not the event organiser " 404 Not Found - "Event not found"
GET	/api/events/organiser/{or ganiserId}	Get all events by a specific organise r	None (Public)	None	200 OK - list of events 404 Not Found - "Organise r not found"

CATEGORY ENDPOINTS					
HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/categories	Get all available categories	None (Public)	None	200 OK - { "items": [{ "id", "categoryName", "description", "defaultDistance" }] }
GET	/api/categories/{id}	Get specific category by ID	None (Public)	None	200 OK - category details 404 Not Found - "Category not found"
POST	/api/categories	Create a new category	Organiser	{ "categoryName": "string", "description": "string", "defaultDistance": 10.0 }	201 Created - category details 400 Bad Request - validation errors 409 Conflict - "Category name already exists"
PUT	/api/categories/{id}	Update an existing category	Organiser	{ "categoryName": "string", "description": "string", "defaultDistance": 10.0 }	200 OK - updated category 403 Forbidden - "Insufficient permissions" 404 Not Found - "Category not found"
DELETE	/api/categories/{id}	Delete a category	Organiser	None	204 No Content 403 Forbidden - "Insufficient permissions" 409 Conflict - "Category is in use by events"

EVENT CATEGORY MANAGEMENT ENDPOINTS					
HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/events/{eventId}/categories	Get all categories for a specific event	None (Public)	None	200 OK - list of event categories with pricing and participant counts 404 Not Found - "Event not found"
POST	/api/events/{eventId}/categories	Add a category to an event	Organiser	{ "categoryId": 1, "entryFee": 150.00, "maxParticipants": 500 }	201 Created - event category details 400 Bad Request - "Category already added" 404 Not Found - "Event or category not found"
DELETE	/api/event-categories/{eventCategoryId}	Remove a category from an event	Organiser	None	204 No Content 404 Not Found - "Event category not found" 409 Conflict - "Cannot"

					remove category with active enrolments"
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ENROLMENT ENDPOINTS					
HTT P Met hod	Route	Descri ption	Role Requir ed	Request Body	Expected Response
GET	/api/enrolments	Get all enrolments with filtering	Any (Logged in)	None (Query: ?eventId=&participantId=&status=)	200 OK - list of enrolments (organisers see all, participants see only theirs) 401 Unauthorized - "Not authenticated"
GET	/api/enrolments/{id}	Get specific enrolment by ID	Any (Logged in)	None	200 OK - enrolment details with event and participant info 403 Forbidden - "Not authorised" 404 Not Found -

					"Enrolment not found"
POST	/api/events/{eventId}/enrol	Enrol a participant in an event	Participant	{ "categoryId": 1 }	201 Created - { "enrolmentId", "eventId", "categoryName", "bibNumber", "status", "paymentsStatus" } 400 Bad Request - "Already enrolled" or "Category is full" 404 Not Found - "Event or category not found"
PUT	/api/enrolments/{id}	Update enrolment status	Participant	{ "status": "Cancelled" }	200 OK - updated enrolment 403 Forbidden - "Not the participant" 404 Not Found - "Enrolment not found"
DELETE	/api/enrolments/{id}	Cancel an enrolment	Participant	None	204 No Content 403 Forbidden

					- "Not the participant" 404 Not Found - "Enrolment not found"
GET	/api/participants/{participantId}/enrolments	Get all enrolments for a participant	Any (Logged in)	None	200 OK - list of participant's enrolments 403 Forbidden - "Not authorised" 404 Not Found - "Participant not found"

RESULT ENDPOINTS					
HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/events/{eventId}/results	Get all results for an event	None (Public)	None (Query: ?categoryId=&page=1&pageSize=20)	200 OK - paginated list of results with participant info 404 Not Found - "Event not found"

GET	/api/results/{id}	Get specific result by ID	Any (Logged in)	None	200 OK - result details 403 Forbidden - "Not authorised" 404 Not Found - "Result not found"
GET	/api/participants/{participantId}/results	Get all results for a participant	Any (Logged in)	None	200 OK - list of participant's results 403 Forbidden - "Not authorised" 404 Not Found - "Participant not found"
POST	/api/enrolments/{enrolmentId}/results	Submit a result for an enrolment	Organiser	{ "finishTime": "01:30:45", "position": 25, "overallRank": 25, "categoryRank": 10, "pacePerKm": 5.45 }	201 Created - result details 400 Bad Request - "Result already exists" 403 Forbidden - "Not an organiser"

					" 404 Not Found - "Enrolment not found"
PUT	/api/results/{id}	Update a result	Organiser	{ "finishTime": "01:30:45", "position": 25, "overallRank": 25, "categoryRank": 10, "pacePerKm": 5.45, "verified": true }	200 OK - updated result 403 Forbidden - "Not an organiser" 404 Not Found - "Result not found"
DELETE	/api/results/{id}	Delete a result	Organiser	None	204 No Content 403 Forbidden - "Not an organiser" 404 Not Found - "Result not found"

WEATHER ENDPOINTS (for Part 3)					
HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/events/{eventId}/weather	Get weather forecast	None (Public)	None	200 OK - list of weather

		for an event			forecasts 404 Not Found - "Event not found"
POST	/api/events/{eventId}/weather	Add or update weather forecast	Organiser	{ "forecastDate": "2026-01-01", "temperature": 18.5, "condition": "Sunny", "humidity": 60, "windSpeed": 12.5, "precipitation": 0 }	201 Created - weather data 403 Forbidden - "Not the event organiser"

DASHBOARD ENDPOINTS					
HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/dashboard/organiser/stats	Get organiser dashboard statistics	Organiser	None	200 OK - { "totalEvents": 5, "totalEnrolments": 250, "upcomingEvents": 3, "participantsCount": 150 }
GET	/api/dashboard/participant/stats	Get participant dashboard statistics	Participant	None	200 OK - { "enrolledEvents": 4, "upcomingEvents": 2, "completedEvents": 2, "personalBests": { "fastest5km": "00:25:30",

					"fastest10km": "00:55:45" } }
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